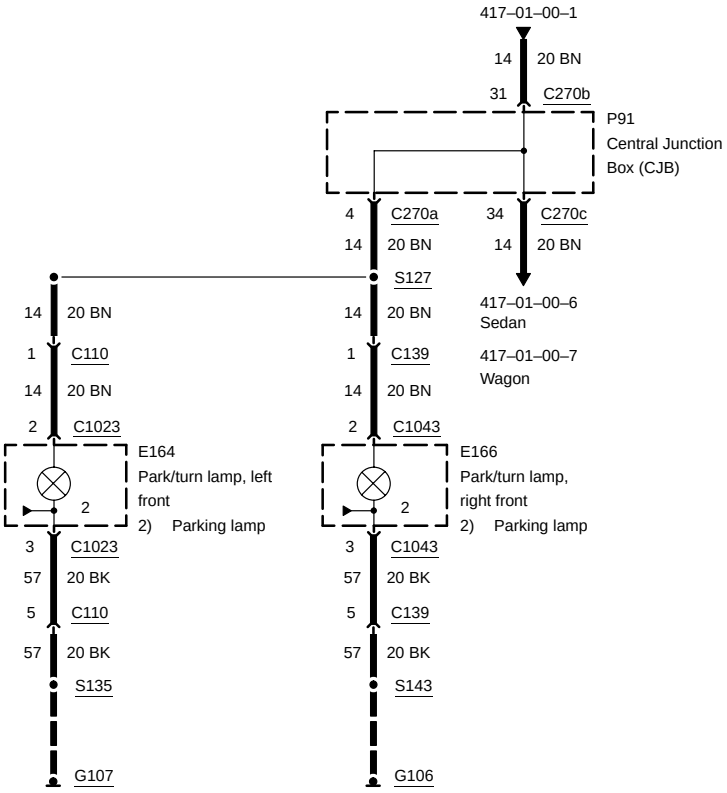without autolamps -4